Fiscal Year:	FY 2026	**Task Last Updated:**	FY 08/14/2025
PI Name:	Gilroy, Simon Ph.D.		
Project Title:	BRIC: Exploring Spaceflight-Linked Changes in Plant Defense Capabilities		
Division Name:	Space Biology		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:			
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	(1) Cell & Molecular Biology (2) Microbiology (3) Plant Biology		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	(1) Translational (Countermeasure) Potential (2) Bioregenerative Life Support		
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Comments:	NOTE: PI formerly at Pennsylvania State University; moved to University of Wisconsin-Madison in 2007 (Info received 7/2009)		
Project Type:	Flight	**Solicitation / Funding Source:**	2016-17 Space Biology (ROSBio) NNH16ZTT001N-FG. App G: Flight and Ground Space Biology Research
Start Date:	10/26/2018	**End Date:**	10/25/2026
No. of Post Docs:	3	**No. of PhD Degrees:**	1
No. of PhD Candidates:	1	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	3
No. of Bachelor's Candidates:	6	**Monitoring Center:**	NASA KSC
Contact Monitor:	Massa, Gioia	**Contact Phone:**	321-861-2938
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Flight Program:	ISS		
Flight Assignment:	NOTE: End date changed to 10/25/2026 per the PI (Ed., 1/3/26) NOTE: End date changed to 10/25/2025 per the PI (Ed., 6/4/25) NOTE: End date changed to 10/25/2024 per NSSC information (Ed., 10/16/23) NOTE: End date changed to 10/25/2023 per NSSC information (Ed., 11/3/22) NOTE: End date changed to 10/25/2022 per NSSC information (Ed., 1/4/22) NOTE: End date changed to 10/25/2021 per G. Massa/KSC; forthcoming in NSSC (Ed., 9/24/20)		
Key Personnel Changes/Previous PI:	None		
COI Name (Institution):	Swanson, Sarah Ph.D. (University of Wisconsin, Madison)		
Grant/Contract No.:	80NSSC19K0126		

Performance Goal No.:	
Performance Goal Text:	
Task Description:	Analysis of the plant biology transcriptomics and proteomics data in the GeneLab data system reveals that a common feature of spaceflight is alteration in plant defense-related transcripts and proteins in space. Anecdotally, plants have also been seen to suffer from pathogen attack during spaceflight missions. Taken together, these observations suggest a hypothesis that plant defense responses may be compromised by spaceflight. Such a disruption of defense would have important implications for the management of plants grown as part of an astronaut life support system. This is especially relevant given the large microbial community on board the International Space Station (ISS) and the possibility that spaceflight triggers enhanced microbial pathogenicity. However, the critical unanswered question is whether the altered defense gene expression seen in spaceflight truly plays out as altered susceptibility to pathogens. Use of the Biological Research in Canisters-Light Emitting Diodes (BRIC-LED) hardware growing Arabidopsis seedlings allows direct testing of this idea. Specifically, the project will: (1) Capitalize on the unique capabilities of the BRIC-LED hardware to analyze the transcriptome and cell wall structure responses of Arabidopsis seedlings grown in spaceflight. Plants will be grown in the hardware and then challenged with the bacterial elicitor peptide flg22. Comparison of subsequent on-orbit responses in wild-type and mutants in the defense regulators AtRBOHD and CML24 will then be made. (2) Use comparisons to ground-based control experiments and clinostat and hypoxia treatment to ask whether such changes are likely linked to the microgravity or hypoxic elements of spaceflight and whether these changes may affect the ability of the plant to defend itself against pathogen challenge. The BRIC-LED/Petri Dish Fixation Unit (PDFU) hardware is well-suited to this experimental approach. Thus, the two-stage PDFU hardware will allow both stimulation (injection of flg22 elicitor onto the seedlings) and subsequent fixation of samples (with RNAlater). As the experiment would use flg22 peptide to elicit defense response, potential problems with flying live pathogen will be circumvented. The experimental design will capitalize on the minimal crew time needed for a BRIC-based experiment and the unique capabilities of the BRIC hardware to treat plants in situ on orbit. The results from this experiment will allow direct testing of the critical question of whether the ability of plants to mount a defense response is compromised during spaceflight.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	This research is addressing how pathogen sensing and response operates in plants, targeting the Ca2+-dependent and reactive oxygen signaling machinery. The results from this work are not only providing insight into how pathogen signals impact plant growth and defense responses in space but also are contributing to our understanding of similar responses on Earth. Pathogen attack is a major limit on crop productivity with estimates of global losses in the range of 20% across multiple major crops such as wheat, maize, rice, soybean, and potato. The pathogen sensing and response systems triggered by the molecular signals of bacterial attack in space are also those employed by plants challenged by pathogens on Earth. Thus, this work aimed at analysis of how reactive oxygen species and Ca2+ signaling is involved in the defense response systems induced during spaceflight should also help our understanding for these same systems in this general plant defense response system on Earth. These results will provide information to highlight novel molecular targets for potential manipulation to tailor plant responses to the wide spectrum of pathogens that attack plants.
Task Progress:	The flight component of this project is awaiting the availability of the Biological Research in Canisters-Light Emitting Diodes (BRIC-LED) hardware and an associated flight opportunity. Fresh seed stocks have been maintained and validated for all the genotypes for the experiment and are regularly being tested for germination and growth characteristics to be ready for a potential flight opportunity. Work has continued to characterize how spaceflight conditions such as hypoxic stress might be influencing plant defense capabilities. Flooding stress and hypoxia are thought to be important elements of the plant spaceflight environment due to the loss of both gravity-driven water flow and buoyancy-driven convection in the microgravity of space. On Earth, flooding stress is thought to lead to shifts in defense capabilities against pathogens, priming the plant to defend itself more effectively once the flooding stress has subsided. Additionally, flooding is thought to suppress responses to herbivory and mechanical damage. Previous spaceflight data from the Advanced Plant Experiments-05 (APEX-05) experiment have been analyzed and combined with ground-based studies. Previous work has shown that when the CAX2 vacuolar Ca2+ transporter is knocked out, plants become resistant to flooding stress on Earth. These same CAX2 knockouts were grown on the International Space Station (the APEX-05 flight experiment). Analysis of patterns of gene expression from the APEX-05 data indicated that the wild type plants exhibited molecular signatures of hypoxia and oxidative stress while growing in space. These changes were coupled to alterations in patterns of gene expression associated with response to both pathogen-attack and physical damage, or herbivory effects that were not seen in the cax2 mutant. Ground-based experiments showed that a hypoxic stress of ~6% oxygen can mimic many of these spaceflight-related events. An additional major effect of spaceflight on the cax2 mutant was a large-scale shift in patterns of gene expression towards those related to cell wall biogenesis, maintenance, and remodeling in the shoot. Glycome profiling was therefore used to catalog cell wall polymer composition. Glycomics uses a suite of monoclonal antibodies to quantify the non-cellulosic cell wall polymer composition through an enzyme-linked immunosorbent assay (ELISA)-based assay. This profiling revealed spaceflight-triggered increases in wall polymers such as xyloglucans and pectins in both wild type and the cax2 mutant. However, a notable difference between the two genotypes was that, in wild type, spaceflight induced upregulation of the xylan components of the wall in only roots; whereas in the cax2 mutant, this change was shifted to being exclusively in the shoot tissues. The xylan content of the wall plays a key role in both structural support and in maintaining a resilient physical barrier for plant defense, indicating a likely major shift in these features is being elicited by growth in space. These results further suggest that the regulatory networks governing these changes to the wall are strongly influenced by the action of the CAX2 gene, potentially through its role in Ca2+-based signaling of hypoxic stress.
Bibliography Type:	Description: (Last Updated: 02/02/2026)

Articles in Peer-reviewed Journals	Roeder A, Shi Y, Yang S, Abbas M, Sasidharan R, Yanovsky M, Casal J, Ruffel S, von Wirén N, Assmann S, Kinscherf N, Bakshi A, Alptekin B, Gilroy S, SharathKumar M, Prat S, Argueso C. "Translational insights into abiotic interactions: From Arabidopsis to crop plants." The Plant Cell. 2025 Jul 1;37(7):koaf140. https://doi.org/10.1093/plcell/koaf140 ; PubMed PMID: 40579370; PubMed Central PMCID: PMC12292048 , Jul-2025
Articles in Peer-reviewed Journals	Johns S, Wiegman E, Bakshi A, Gilroy S. "The cyclic nucleotide gated channels CNGC2 and CNGC4 support systemic wound responses in Arabidopsis thaliana." Front. Plant Sci. 2025 Aug 21;16:1545065. https://doi.org/10.3389/fpls.2025.1545065 ; PMID: 40918972; PMCID: PMC12411437 , Aug-2025
Articles in Peer-reviewed Journals	Seo D, Strickland HF, Zhou M, Barker R, Ferl RJ, Paul AL, Gilroy S. "GLARE: Discovering hidden patterns in spaceflight transcriptome using representation learning." npj Microgravity. 2025 Oct 28;11:76. https://doi.org/10.1038/s41526-025-00525-5 ; PMID: 41152268; PMCID: PMC12569054 , Oct-2025
Articles in Peer-reviewed Journals	Bakshi A, Gilroy S. "Calcium signaling in hypoxic response." Plant Physiol. 2024 Dec 24;197(1):kiae654. https://doi.org/10.1093/plphys/kiae654 ; PMID: 39707915 , Dec-2024